Nguyen Xuan Cuong

Da Nang, Vietnam | +84 327 755 183 | nxcforwork@gmail.com
GitHub: nxc1802 | LinkedIn: cuong-nguyen-2715a136b | Portfolio: nxc-s-portfolio.vercel.app

Summary

Senior AI & Software Engineer with 5+ years of hands-on experience in AI and software engineering, specializing in building production-grade Generative AI agents, Advanced RAG architectures, and real-time Computer Vision systems. Proven track record in bridging deep learning research with scalable full-stack and cloud deployments (AWS, GCP, Azure, Hugging Face, Vercel, Supabase, Nosana). Adept at technical leadership, architectural design, and rapid prototyping using modern AI-assisted engineering workflows. Fluent in English (IELTS 6.5 equivalent) for international technical collaboration and research.

Education

FPT University – Da Nang, Vietnam
B.Sc. in Information Technology (AI Major) | GPA: 8.28/10.00 | *Recipient of 100% Full-tuition FPT University Scholarship.*

Work Experience

PiXerse Technology – Da Nang, Vietnam Mar 2025 – Present
Senior AI Developer / AI Architect

- Architected 'Pixity', a real-time AI tourism agent ecosystem integrating transport, weather, and event APIs; won 2nd Runner-up at GDG DevFest MienTrung 2025.
- Designed Graph-RAG pipeline combining Neo4j graph databases and Pinecone vector search for spatial-temporal itinerary planning.
- Implemented multi-turn conversational agents with autonomous tool-calling capabilities using Gemini and OpenRouter APIs.
- Built scalable asynchronous backend microservices using FastAPI and Node.js; optimized containerized CI/CD workflows with Docker.

Arcronics Solution – Da Nang, Vietnam Jun 2024 – Feb 2025
Senior AI Engineer / Embedded AI Specialist

- Researched and trained deep learning models for malware classification and cyber attack-type detection.
- Quantized, pruned, and integrated lightweight AI models into hardware systems and microchips (Firmware, FPGA).
- Built automated MLOps pipelines for data preprocessing, feature extraction, model retraining, and continuous evaluation.

Honors & Awards

2nd Runner-up: Hackathon GDG DevFest MienTrung 2025. | **Second Prize**: AI4SE 2025 Competition.
4th Place: Hackathon Nosana Builders Challenge: Agents 102. | **Superteam Honorable Mention**: Timelocked Wallet (2025).
Finalist: AI4LIFE 2024, Research Connection 2025, Research Festival 2025, and Grab The Beyond 2026.

Research & Publications

MAFIS: Multi-Agent Imitation by learning and sampling from FactorIzed Soft Q-function.
Identification of Plant Leaf Diseases: Using CNN and SVM for precision agriculture.
The GYM Exercise Classification Dataset: 2D/3D pose analysis using CNN-LSTM and Mediapipe.
Towards Sparse Adversarial Perturbations: Gradient-Guided Approach for Efficient Image Attacks.

Projects

Manim Agent – Distributed Educational AI Storyboard & Video Platform Oct 2023 – May 2024
Role: Lead AI & Fullstack Architect / Team Size: 2 / Layers: FastAPI, Celery, Redis AOF, Docker Sandbox, React, Supabase

- Architected an end-to-end multi-service AI system that transforms educational ideas into structured storyboards, generates executable Manim Python scripts, and renders HD mathematical videos.
- Implemented Human-in-the-Loop review flow via WebSockets and real-time state synchronization for scene-by-scene video previews.
- Isolated untrusted code execution using asynchronous Celery worker queues deployed within a protected Hugging Face Docker Space runtime.

MathSolver – Multi-Agent Geometry Problem Solver & Motion Renderer Feb 2023 – Sep 2023
Role: Senior AI Engineer / Team Size: 3 / Layers: Multi-Agent AI System, OCR Engine, Manim Engine, FastAPI, Next.js

- Developed a multi-agent system extracting geometric shapes and equations from images via OCR and generating step-by-step mathematical reasoning paths.

- Designed automated code generation pipeline translating proof steps into 2D/3D Manim motion graphics with multi-LLM fallback orchestration over OpenRouter APIs.
- Containerized backend services with Docker and Cairo/Pango system bindings to deliver real-time video compilation.

Medagen – AI Patient Triage & Diagnostic Assistant Jul 2022 – Jan 2023
Role: Fullstack AI Developer / Team Size: 6 / Layers: LangChain ReAct Agents, Advanced RAG (Supabase pgvector/Qdrant), Computer Vision

- Built a clinical diagnostic assistant utilizing LangChain ReAct Agent framework and Gemini for multi-turn symptom triage, differential diagnosis, and medical tool-calling.
- Architected an Advanced RAG system over medical knowledge bases using 768-dim vector embeddings (Supabase Vector / Qdrant), hybrid search, and semantic document chunking.
- Integrated Computer Vision models (Vision Transformers / custom CNNs) for dermatological lesion classification and multimodal symptom verification.

MagicCV – Autonomous Resume Tailoring AI Agent Jan 2022 – Jun 2022
Role: AI Engineer / Team Size: 2 / Layers: Mastra Agent Framework, Supabase pgvector, LLM Ranking Engine, pdflatex Compiler

- Created an autonomous CV tailoring agent (Nosana Builders Challenge finalist) that extracts candidate profiles from GitHub, LinkedIn, and YouTube APIs.
- Implemented 768-dim vector similarity search with 3-level fallback strategies and Gemini LLM ranking algorithms to match career data with job descriptions in 3 seconds.
- Rendered publication-quality PDFs via Nunjucks LaTeX templating; containerized with Docker and deployed to Nosana decentralized GPU compute network.

Lab Management – Real-Time AI Computer Vision Dashboard May 2021 – Dec 2021
Role: Computer Vision & Software Engineer / Team Size: 5 / Layers: YOLOv8, ByteTrack, Janus WebRTC, WebSockets, Next.js

- Deployed YOLO object detection and ByteTrack multi-object tracking models for real-time indoor occupancy tracking and crowd flow analytics.
- Streamed live video telemetry with Janus WebRTC servers and WebSockets, delivering low-latency spatial tracking analytics to Next.js dashboards.

Skills

Languages: Python (4-Applied theory, 5+ yrs), C++ (4-Applied theory, 5 yrs), Java (3-Practical application, 3+ yrs), TypeScript / JavaScript (3-Practical application, 3+ yrs), SQL (3-Practical application, 4+ yrs), Shell Script
Language Proficiency: English (IELTS 6.5 Equivalent – Professional working proficiency), Vietnamese (Native)
AI, ML, Deep Learning & CV: Machine Learning (5-Recognized authority, 3+ yrs), Deep Learning (5-Recognized authority, 3+ yrs – PyTorch, TensorFlow), Computer Vision (5-Recognized authority, 3+ yrs – YOLOv8/10, MediaPipe, ByteTrack, ViT), NLP (3-Practical application, 3+ yrs)
Generative AI & Agent Frameworks: LangChain, LangGraph, Mastra, Model Context Protocol (MCP), ReAct Agents, Tool Calling, Advanced Graph-RAG, Neo4j, Vector DBs (Qdrant, Pinecone, Supabase pgvector), 768-dim Embeddings
Backend & Frontend: FastAPI (4-Applied theory, 4+ yrs), Node.js (3-Practical application, 3+ yrs), ReactJS / Next.js (3-Practical application, 3+ yrs), Redis (AOF), Celery, WebSockets, Janus WebRTC, PostgreSQL / Supabase
DevOps & Cloud Deployment: Docker & Docker Compose (3-Practical application, 3+ yrs), Cloud Platforms (AWS, GCP, Azure, Hugging Face Spaces, Vercel, Nosana) (3-Practical application, 3+ yrs), Git (GitHub, GitLab)